

Wenduo Cheng

Carnegie Mellon University, Pittsburgh, PA, United States
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Education

Carnegie Mellon University

PH.D. in Computational Biology

- GPA: 4.00/4.33
- Advisor: Jian Ma

Pittsburgh, PA
Aug 2024 - Present

Carnegie Mellon University

M.S. in Computational Biology

- GPA: 4.06/4.33
- Thesis: Leveraging pre-trained language models to address genomics tasks
- Awards: Research Excellence and Honors; Outstanding Academic Achievement

Pittsburgh, PA
Aug 2022 - May 2024

Duke Kunshan University / Duke University

B.S. in Biology / B.S. in Interdisciplinary Studies

- GPA: 3.9/4.0, *Magna Cum Laude*

Kunshan, China / Durham, NC
May 2018 - May 2022

Research Interests

Genomics Foundation Models; AI Agents; Autonomous Laboratory

Publications

Preprints

Towards Human-Led, Agent-Driven Autonomous Laboratories for the Life Sciences [paper]

Wenduo Cheng, Mingqian Ma, Shuaike Shen, Anna Hupalowska, Jennifer E. Rood, Yang Zhang, Gaurav Agrawal, Christine Bakan, Michelle A. Lee, Aviv Regev, Jian Ma

In: [Preprints.org](#) 2026.

AgentCo-op: Retrieval-Based Synthesis of Interoperable Multi-Agent Workflows [paper] [code] [website]

Shuaike Shen[†], Wenduo Cheng[†], Shike Wang, Mingqian Ma, Jian Ma

In: [arXiv](#) arXiv:2605.20425. 2026.

Peer-Reviewed Articles

Phase response curve and RNA-sequencing demonstrate spiders' sensitivity to light and pinpoint candidate light-responsive genes [paper]

Natalia Toporikova, Wenduo Cheng, Leyuan Qian, Andrew Mah, Thomas Clarke, Thomas C. Jones, Darrell Moore, Nadia A. Ayoub

In: [PLOS ONE](#) 21(9), e0344063. 2026.

SKILLFOUNDRY: Building Self-Evolving Agent Skill Libraries from Heterogeneous Scientific Resources

[paper] [code] [website]

Shuaike Shen[†], Wenduo Cheng[†], Mingqian Ma, Alistair Turcan, Martin Jinze Zhang, Jian Ma

In: [Conference on Language Modeling \(COLM\)](#) 2026.

DNALONGBENCH: A Benchmark Suite for Long-Range DNA Prediction Tasks [paper] [code]

Wenduo Cheng[†], Zhenqiao Song[†], Yang Zhang[†], Shike Wang, Danqing Wang, Muyu Yang, Lei Li, Jian Ma

In: [Nature Communications](#) 16, p. 10108. 2025. ([Editors' Highlights](#))

L2G: Repurposing Language Models for Genomics Tasks [paper] [code]

Wenduo Cheng, Junhong Shen, Mikhail Khodak, Jian Ma, Ameet Talwalkar

In: [Transactions on Machine Learning Research \(TMLR\)](#) 2025.

Specialized Foundation Models Struggle to Beat Supervised Baselines [paper] [code]

Zongzhe Xu, Ritvik Gupta, Wenduo Cheng, Alexander Shen, Junhong Shen, Ameet Talwalkar, Mikhail Khodak

In: [International Conference on Learning Representations \(ICLR\)](#) 2025.

The Special and General Mechanism of Cyanobacterial Harmful Algal Blooms [paper]

Wenduo Cheng[†], Somin Hwang[†], Qisen Guo, Leyuan Qian, Weile Liu, Yang Yu, Li Liu, Yi Tao, Huansheng Cao

In: [Microorganisms](#) 11(4), p. 987. 2023.

Draft Genome Sequence of an Epibiotic Bacterium, *Bacillus cereus*, Isolated from Cyanobacterial Blooms in Lake Taihu, China [paper]

Xiaoyuan Chen, Yinuo Yang, Yang Yu, Qisen Guo, Somin Hwang, **Wenduo Cheng**, Huansheng Cao
In: **Microbiology Resource Announcements** 12(3), e00936–22. 2023.

Cyanobacterial Blooms Are Not a Result of Positive Selection by Freshwater Eutrophication [paper]

Yang Yu[†], **Wenduo Cheng**[†], Xiaoyuan Chen, Qisen Guo, Huansheng Cao
In: **Microbiology Spectrum** 10(6), e03194–22. 2022.

([†] Equal contribution)

Professional Experience

Genentech – Research Intern

Advisors: Bo Li and Monica Ge

South San Francisco, CA

May 2026 - Aug 2026

- Developed OPSAgent, a specialized agent that automates optical pooled screening data analysis.

Research Experience

Jian Ma’s Group at Carnegie Mellon University

Advisor: Jian Ma

Pittsburgh, PA

Jan 2023 – Present

- Ph.D. research on developing and evaluating biological foundation models and autonomous agents for genomics, spatial transcriptomics, and biological discovery.

The Laboratory of Environmental Sciences at Duke Kunshan University

Advisor: Huansheng Cao

Kunshan, China

Dec 2020 - May 2022

- **Project 1: Mechanisms of Cyanobacterial Harmful Algal Blooms.** Synthesized evidence on how cyanobacterial traits and elevated nutrients jointly drive harmful algal blooms (published in *Microorganisms*).
- **Project 2: Positive Selection in Cyanobacterial Algal Blooms.** Investigated the role of positive selection in cyanobacterial bloom formation using molecular evolutionary analyses, pangenomics, and metatranscriptomics (published in *Microbiology Spectrum*).
- **Project 3: Draft Genome Assembly of Bacteria from Lake Taihu.** Isolated and sequenced a bacterial strain from Lake Taihu, then assembled and annotated its draft genome (published in *Microbiology Resource Announcements*).

The Statistical Genetics Laboratory at Westlake University

Advisor: Jian Yang

Hangzhou, China

Jul 2021 - Sep 2021

- Built a variant-calling pipeline for single-cell RNA-seq, bulk RNA-seq, and DNA-seq data and demonstrated SNP calling and genotyping from 10x Genomics scRNA-seq.

Natalia Toporikova’s Group at Washington and Lee University

Advisor: Natalia Toporikova

Remote

May 2020 - Jun 2021

- Analyzed spider RNA-seq responses to light pulses and collection time, identifying 528 differentially expressed transcripts with reversed expression patterns as candidate light-sensitive circadian genes (published in *PLOS ONE*).

Professional Service

Journal Reviewer Nature Biotechnology, Nature Machine Intelligence

Conference Reviewer RECOMB 2024-2026, ACM BCB 2025, ISMB/ECCB 2025

Teaching Assistant Computational Medicine (CMU 02718, 23 fall), Computational Genomics (CMU 02710, 24 spring)

RELATED COURSEWORK

CS	Deep Learning, Natural Language Processing, Multimodal Machine Learning, Algorithms and Data Structure
Math	Probability and Statistics, Linear Algebra, Multivariable Calculus, Random Variables and Stochastic Processes
Biology	Cell and Molecular Biology, Genetics and Evolution, Biophysics, Biochemistry, Microbiology, Population Genomics, Functional Genomics, Molecular Genetic Analysis
Comp Bio	Computational Genomics, Computational Medicine, Biological Modeling and Simulation, Automation of Scientific Research

Skills

Programming Proficient in Python, R and Bash; Familiar with Java, Golang, MATLAB, Mathematica and SQL

Miscellaneous Git, $\LaTeX$, Linux, Shell, Slurm, PyTorch, Scikit learn, VS Code, Jupyter Notebook and Markdown